Darpan Lunagariya

B.Tech (Honors) – Information and Communication Technology with minor in Computational Science

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Education

B.Tech. (Honors) in ICT with minor in CS (Dhirubhai Ambani University) 2022 - Present
***CPI**: 9.06* *Gandhinagar, Gujarat*

Standard 12 (CVM Higher Secondary Education Complex - GSHSEB) 2022
***Percentage**: 89%* *Anand, Gujarat*

Standard 10 (Gnanyagna Vidhyalaya - GSEB) 2020
***Percentage**: 92%* *Anand, Gujarat*

Experience

Summer Research Internship May 2025 – July 2025
Automatic Printable Shellcode Generation for ARMv7 Architecture *Gandhinagar, Gujarat*

- Developed a tool that automatically converts any arbitrary machine code to printable characters and decodes it to actual machine code at runtime, under the guidance of Prof. Anish Mathuria.
- We are 1st in the world to implement such a tool for ARMv7 architecture.

Projects

Torrent Downloading Client | *Python*

- Built a BitTorrent client that downloads torrent content by establishing connections with multiple peers and trackers.
- Implemented the BitTorrent protocol to retrieve peer information from trackers and download file pieces from peers. Utilized multi-threading and piece hash verification to ensure efficiency, reliability and consistency.

URL Shortner (ShrinkURL) | *Spring Boot, Hibernate, MySQL, Redis, Thymeleaf*

- Built a scalable and efficient URL shortening service with features including stateless login using access and refresh tokens, in-memory caching for faster redirects, rate limiting to prevent abuse and analytics to track click counts.
- Implemented premium user capabilities allowing creation of custom short aliases, along with access to extended expiration controls.

Character Device Driver for Linux Based OS | *C, Shell Script*

- Developed a character device driver for Linux-based systems to facilitate communication with hardware through a file interface at `/dev/pmad0`.
- Designed the driver to directly access physical memory, enabling read and write operations at specific memory locations instead of interacting with standard devices.

Database of A Freelancing Platform | *PostgreSQL*

- Designed and implemented a comprehensive freelancing database system with Entity-Relationship modeling, normalizing to Boyce-Codd Normal Form (BCNF) to eliminate data redundancy and ensure optimal database structure.
- Executed ER-to-relational mapping and deployed the database on PostgreSQL using Data Definition Language (DDL) commands, while developing complex retrieval queries for efficient data access and manipulation.

Technical Skills

Languages: C, C++, Assembly, Python, Java, SQL, MATLAB
Developer Tools: Git, Docker, Shell Scripting, OpenMP, MPI, GDB debugger
Frameworks: SpringBoot, Hibernate
Databases: MongoDB, PostgreSQL, Redis
Soft Skills: Problem Solving, Communication, Time Management, Mentorship
Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Object Oriented Programming, Operating Systems, Distributed Systems, High Performance Computing

Position Of Responsibility

Treasurer
December 2024 - Present

IEEE Student Branch DA-IICT

- Managing finances, budgeting, and overseeing financial transactions for the IEEE Student Branch at DA-IICT.

Teaching Assistant
January 2025 - May 2025

Course: Introduction to Communication Systems (CT216)

- Assisted in conducting laboratory sessions, evaluated student lab performance and project work, resolved academic queries, and graded assignments and final assessments.

Core Member
November 2023, November 2024

i.Fest (Annual Techo-cultural Festival at DA-IICT)

- Led Event Management and Public Relations for straight 2 years in the Gujarat's largest annual techno-cultural festival, i.Fest

Achievements

- NCC (National Cadet Corps) "A" Certificate
- 3rd Rank in JabWeCode (Organized by Ahmedabad University)
- Secured 5th rank in Gujarat State Higher Secondary Board Exams (GSHSEB) – 2022.